Assignment 11: Surviving the Titanic

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Exercise 1

```r
train_df <- read_csv(file = "train.csv")
```

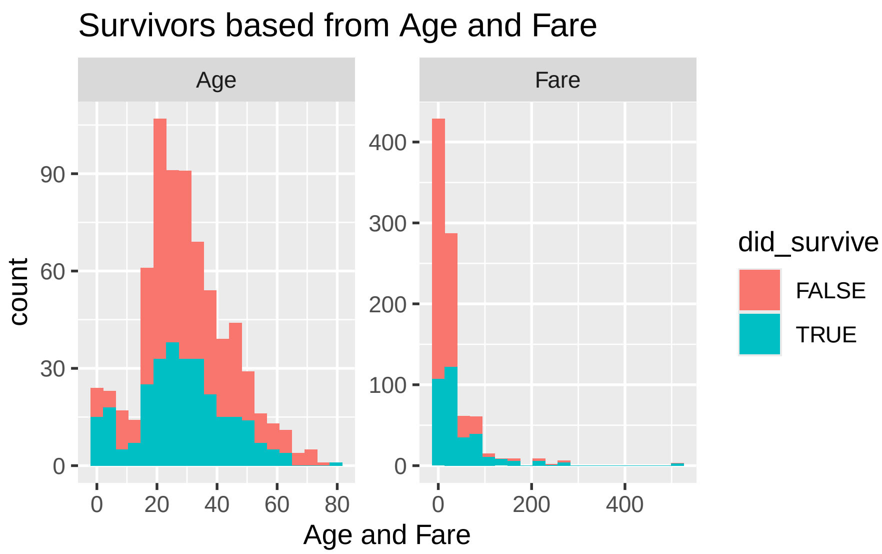
```
## Rows: 891 Columns: 12
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (5): Name, Sex, Ticket, Cabin, Embarked
## dbl (7): PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
train_df <- read_csv(file = "train.csv",
  col_types = cols(
    Pclass = col_character(),
    SibSp = col_character(),
    Parch = col_character()
  ))

train_df<- train_df %>%
  mutate(did_survive = as.logical(Survived))
```

Exercise 2

```r
train_df %>%
  pivot_longer(cols = Age|Fare,
               names_to="name",
               values_to = "value") %>%
  ggplot() +
  geom_histogram(aes(x = value, fill = did_survive), bins = 20) +
  facet_wrap(~name, scales = "free") +
  labs(title = "Survivors based from Age and Fare",
       x = "Age and Fare")
```

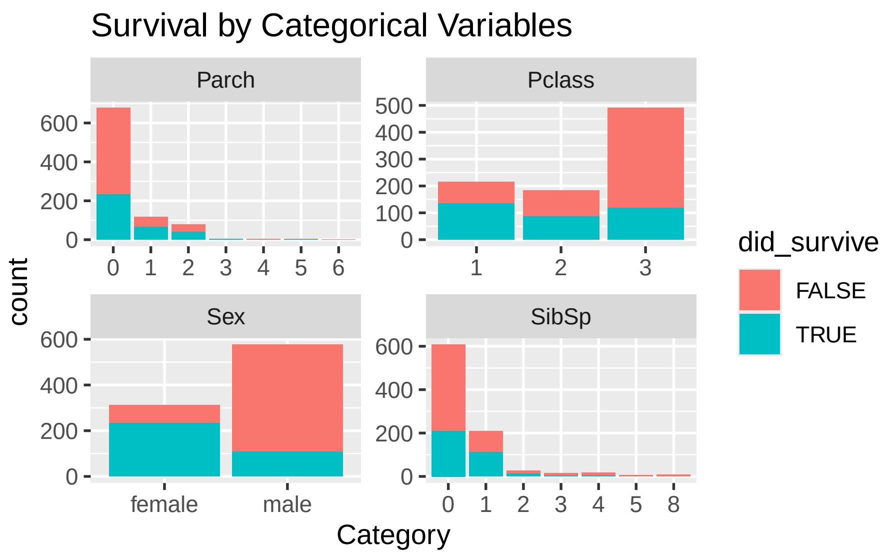


i. Using your graphs, how do the distributions of the Age and Fare variables differ between survivors and non-survivors? Do you think these differences will be helpful for predicting who survived?

The graph shows that younger passengers, especially those under 15, were more likely to survive, while people aged 20-40 had lower survival rates. For fare, those who paid less had a lower chance of survival, while passengers who paid higher fares tended to survive more. This suggests that both age and fare could be useful factors for predicting who survived on the Titanic.

Exercise 3

```
train_df %>%
  pivot_longer(cols = c(Pclass, Sex, Parch, SibSp),
               names_to = "name",
               values_to = "value") %>%
  ggplot() +
  geom_bar(aes(x = value, fill = did_survive)) +
  facet_wrap(~name, scales = "free") +
  labs(title = "Survival by Categorical Variables",
       x = "Category")
```



i. Using your graphs, explain how each of these categorical variables is related to surviving (or not surviving) the Titanic. Do these results make sense given what you know about the Titanic's sinking?

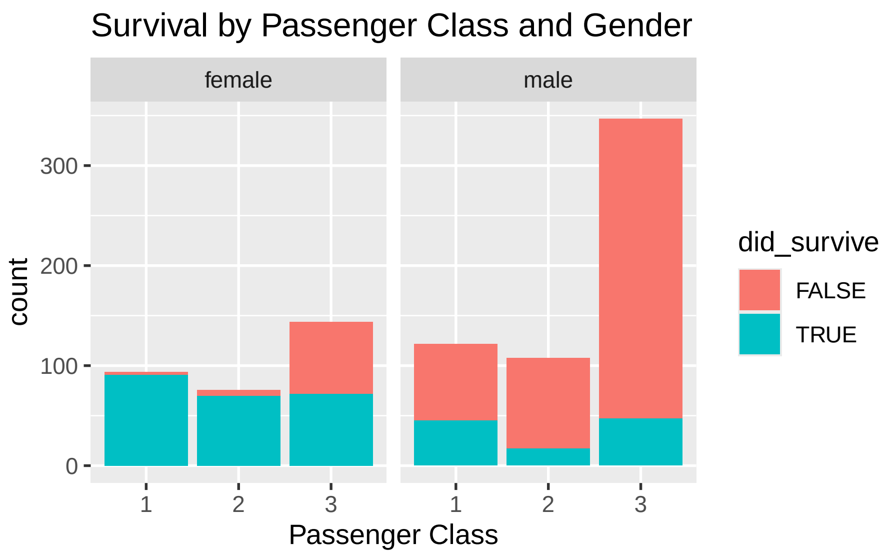
The graphs show that first-class passengers had better survival rates than those in lower classes, probably because they had easier access to lifeboats. People traveling without family (Parch = 0) also survived more, maybe because they could focus on themselves. Women had higher survival rates than men, because to the "women and children first" rule. And people traveling solo (SibSp = 0) seemed to have better odds of getting a lifeboat spot.

ii. Which variables do you think might be most useful for predicting survival, and why?

Sex and Pclass seem like the best predictors for survival since they show clear differences between those who survived and who didn't, fitting with what we know about the Titanic's evacuation.

Exercise 4

```
train_df %>%
  ggplot() +
  geom_bar(aes(x = Pclass, fill = did_survive)) +
  facet_wrap(~Sex) +
  labs(title = "Survival by Passenger Class and Gender",
       x = "Passenger Class")
```



The bar graph shows that survival on the Titanic varied a lot depending on both class and gender. Women had higher survival rates in every class, especially in first and second class. For men, survival rates were much lower overall, with second- and third-class men having the lowest chances. First-class men did have a slightly better chance than men in lower classes, but still much lower than women. This pattern suggests that survival was heavily influenced by both gender and class, probably because evacuation prioritized women and children, and higher-class passengers had better lifeboat access.

Exercise 5

```
train_df %>%
  summarize(
    count = n(),
    missing = sum(is.na(Age)),
    fraction_missing = mean(is.na(Age))
  )
```

count	missing	fraction_missing
891	177	0.1986532

```
train_imputed <- train_df %>%
  mutate(
    age_imputed = if_else(
      condition = is.na(Age),
      true = median(Age, na.rm = TRUE),
      false = Age
    )
  )
```

```
train_imputed %>%
  summarize(
```

```
    count = n(),
    missing = sum(is.na(age_imputed)),
    fraction_missing = mean(is.na(age_imputed))
  )
```

count	missing	fraction_missing
891	0	0

Exercise 6

```
model_1 <- glm(Survived ~ age_imputed,
               data = train_imputed,
               family = "binomial")
```

```
model_1_preds <- train_imputed %>%
  add_predictions(model_1, type = "response") %>%
  mutate(
    outcome = if_else(pred > 0.5, 1, 0)
  )
```

```
model_1_preds %>%
  mutate(
    correct = if_else(
      condition = Survived == outcome,
      true = 1,
      false = 0
    )
  ) %>%
  summarize(
    total_correct = sum(correct),
    accuracy = total_correct / n()
  )
```

total_correct	accuracy
549	0.6161616

i. What is the accuracy of your model? Does this seem good or bad?

The accuracy is 61.6% which is higher than expected because the model is predicting the data.

Exercise 7

```
logistic_cv1 <- cv.glm(train_imputed, model_1, cost, K=5)
```

```
logistic_cv1$delta
```

```
## [1] 0.3838384 0.3838384
```

Exercise 8

```r
model_2 <- glm(Survived ~ age_imputed + SibSp + Pclass + Sex,
               data = train_imputed,
               family = "binomial")
logistic_cv2 <- cv.glm(train_imputed, model_2, cost, K=5)
logistic_cv2$delta
```

```
## [1] 0.1986532 0.2006711
```

```r
model_3 <- glm(Survived ~ age_imputed * Pclass * Sex + SibSp,
               data = train_imputed,
               family = "binomial")
logistic_cv3 <- cv.glm(train_imputed, model_3, cost, K=5)
logistic_cv3$delta
```

```
## [1] 0.1885522 0.1847443
```

Model 3 has the lowest cross-validation error, which makes it the best at predicting survival. This lines up with what we expect, since Model 3 includes interactions between `age_imputed`, `Pclass`, and `Sex`, letting it pick up on more patterns, like how women and kids in higher classes were more likely to survive. Model 2, which uses several predictors but doesn't include these interactions, also does okay, showing that things like class and family size matter for survival. Model 1, which only uses age, has the highest error, so age alone isn't a great predictor. Overall, Model 3 is the strongest, but when models get more complex, they can sometimes fit too closely to one dataset, so it's good to keep things simple where possible.

Academic Integrity statement

No tools other than the teachers guidance from Slack was used to complete this assignment.